

Mathematical Description of the Zeroth Kind of Arithmetic Expression Space ($\mathfrak{E}_0$): Diffusion from a Single Zero Point

Mingli Yuan, Gemini AI

July 28, 2025

Abstract

This note details the mathematical construction of a zeroth kind of Arithmetic Expression Geometry space, denoted $\mathfrak{E}_0$.

$\mathfrak{E}_0$ is characterized by an assignment function a emanating from a single singular point (a zero point) in a hyperbolic plane, modeled here by the Poincaré disk.

Its evolution, under ideal conditions (propagation along the gradient of a), exhibits high symmetry, analogous to a diffusion circle. We explore its geometric properties, including the assignment function itself, characteristic lines (similarity, popularity, additive, and multiplicative lines), and its basic topological nature. This model serves as a foundational case for understanding AEG spaces with point-like singularities, contrasting with the line-like singularities of the first kind space $\mathfrak{E}_1$.

1 Introduction

This note introduces a specific model for a zeroth kind of Arithmetic Expression Geometry (AEG) space, which we will denote by $\mathfrak{E}_0$.

Unlike the first kind of AEG space ($\mathfrak{E}_1$), which is typically built upon the upper half-plane and features a line of zero points (e.g., the imaginary axis for $a(z) = -\text{Re}(z)/\text{Im}(z)$), $\mathfrak{E}_0$ is conceptualized within the Poincaré disk with a single, central zero point acting as a source of 'diffusion' for the assignment function a .

The motivation for studying $\mathfrak{E}_0$ is to understand the behavior of AEG principles in a highly symmetric setting originating from a point singularity. This provides a contrast and a building block for more complex AEG constructions. We assume the curvature of this space is $K = -\lambda^2$, and the assignment function $a(s)$ is derived from the AEG flow equation $da/ds = \sqrt{\mu^2 + (\lambda a)^2}$, where s is the hyperbolic distance in this $K = -\lambda^2$ space.

2 Geometric Setting and Coordinates

The geometric setting for $\mathfrak{E}_0$ is a hyperbolic plane with constant Gaussian curvature $K = -\lambda^2$, where $\lambda > 0$ is the multiplicative parameter from the arithmetic gene (μ, λ) . We model this space using the Poincaré disk $\mathbb{D}^2 = \{w \in \mathbb{C} : |w| < 1\}$.

The center of the disk, $w = 0$, is chosen as the unique zero point of the assignment function a , so $a(w = 0) = 0$. This point acts as the source or singularity.

The path parameter s is the hyperbolic distance from this central zero point $w = 0$ to any point w in the disk. If $r_e = |w|$ is the Euclidean distance from the origin in the Poincaré disk model, the relationship between s and r_e for a space with curvature $K = -\lambda^2$ is:

$$s(r_e) = \frac{2}{\lambda} \operatorname{arctanh}(r_e)$$

This means $\lambda s = 2 \operatorname{arctanh}(r_e)$. As $r_e \rightarrow 1$ (approaching the boundary of the disk), $s \rightarrow \infty$.

3 Assignment Function $a(s)$ and $a(r_e)$

The assignment function $a(s)$ depends only on the hyperbolic distance s from the origin, reflecting the radial symmetry of the construction. It is the solution to the AEG flow equation $\frac{da}{ds} = \sqrt{\mu^2 + (\lambda a)^2}$ with the initial condition $a(s = 0) = 0$. The solution is:

$$a(s) = \frac{\mu}{\lambda} \sinh(\lambda s)$$

where μ is the additive generator strength and λ is the multiplicative generator strength from the arithmetic gene (μ, λ) . We assume $\mu > 0$ and $\lambda > 0$.

Substituting $s(r_e) = \frac{2}{\lambda} \operatorname{arctanh}(r_e)$ into $a(s)$, we get a as a function of the Euclidean radius r_e :

$$\lambda s = \lambda \left(\frac{2}{\lambda} \operatorname{arctanh}(r_e) \right) = 2 \operatorname{arctanh}(r_e)$$

So,

$$a(r_e) = \frac{\mu}{\lambda} \sinh(2 \operatorname{arctanh}(r_e))$$

Using the identity $\sinh(2y) = \frac{2 \tanh y}{1 - \tanh^2 y}$, and letting $y = \operatorname{arctanh}(r_e)$ (so $\tanh y = r_e$), we have:

$$\sinh(2 \operatorname{arctanh}(r_e)) = \frac{2r_e}{1 - r_e^2}$$

Thus, the assignment function in terms of Euclidean radius is:

$$a(r_e) = \frac{\mu}{\lambda} \left(\frac{2r_e}{1 - r_e^2} \right)$$

This function is zero at $r_e = 0$ and tends to infinity as $r_e \rightarrow 1$.

4 Characteristic Geometric Lines

The structure of $\mathfrak{E}_0$ can be further understood by examining its characteristic lines.

4.1 Contour Lines of $a(r_e)$ (Similarity Lines)

Contour lines are defined by $a(r_e) = \text{constant}$. Since $a(r_e)$ depends only on r_e (for fixed μ, λ), and $\frac{2r_e}{1-r_e^2}$ is monotonic for $r_e \in [0, 1)$, the contour lines are curves where $r_e = \text{constant}$. These are Euclidean circles centered at the origin $w = 0$ in the Poincaré disk model. They represent states with the same "process richness" or assignment value.

4.2 Gradient Lines of $a(r_e)$ (Popularity Lines)

The gradient $\nabla a(r_e)$ indicates the direction of the most rapid change in a . Due to the radial symmetry, $\nabla a(r_e)$ points radially outward (or inward if $\mu/\lambda < 0$, but we assume $\mu/\lambda > 0$). These gradient lines are Euclidean straight line segments passing through the origin (radii of the disk).

4.3 Additive and Multiplicative Lines

The directions of additive and multiplicative lines are determined by the angle $\alpha(a)$, defined as $\alpha(a) = \arctan\left(\frac{\mu}{\lambda a}\right)$. Substituting $a(r_e) = \frac{\mu}{\lambda} \left(\frac{2r_e}{1-r_e^2}\right)$:

$$\lambda a(r_e) = \mu \left(\frac{2r_e}{1-r_e^2}\right)$$

So,

$$\alpha(r_e) = \arctan\left(\frac{\mu}{\mu \left(\frac{2r_e}{1-r_e^2}\right)}\right) = \arctan\left(\frac{1-r_e^2}{2r_e}\right)$$

This angle $\alpha(r_e)$ remarkably depends only on the Euclidean radius r_e and is independent of μ and λ . Let $\vec{g}$ be the unit radial vector (gradient direction) and $\vec{c}$ be the unit tangential vector (contour direction).

- **Additive Line Direction** ($\vec{d}_{\text{add}}$): $\vec{d}_{\text{add}} \propto \cos(\alpha(r_e))\vec{g} + \sin(\alpha(r_e))\vec{c}$.
- **Multiplicative Line Direction** ($\vec{d}_{\text{mult}}$): $\vec{d}_{\text{mult}} \propto -\sin(\alpha(r_e))\vec{g} + \cos(\alpha(r_e))\vec{c}$ (orthogonal to $\vec{d}_{\text{add}}$).

Behavior of $\alpha(r_e)$:

- Near the zero point ($r_e \rightarrow 0$): $2r_e \rightarrow 0^+$, $1 - r_e^2 \rightarrow 1$. So, $\alpha(r_e) \rightarrow \arctan(+\infty) = \pi/2$. Additive lines are primarily tangential (circulating the origin), and multiplicative lines are primarily radial.

- Near the boundary ($r_e \rightarrow 1$): $1 - r_e^2 \rightarrow 0^+$, $2r_e \rightarrow 2$. So, $\alpha(r_e) \rightarrow \arctan(0^+) = 0$. Additive lines are primarily radial, and multiplicative lines are primarily tangential.

The zero point $w = 0$ acts as an "additive singularity" (source of spiraling additive lines), while the boundary $r_e = 1$ acts as an "effective multiplicative singularity" where multiplicative behavior (lines becoming parallel to contours) dominates.

5 Topological Property

If we consider the space $\mathfrak{E}_0$ with its central singularity (the zero point $w = 0$) removed, i.e., $\mathbb{D}^2 \setminus \{0\}$, its first homology group is $H_1(\mathbb{D}^2 \setminus \{0\}) \cong \mathbb{Z}$. This indicates the presence of non-trivial loops that can be drawn around the singularity. This is analogous to a cylinder or a punctured hyperbolic plane. This topological feature may be relevant for understanding global properties such as arithmetic torsion when mapping to/from the Accumulative Commutative Space (ACS) for paths that encircle this singularity.

6 Conclusion

The Zeroth Kind of Arithmetic Expression Space, $\mathfrak{E}_0$, defined by a single dif-fusing zero point in a hyperbolic plane of curvature $K = -\lambda^2$, exhibits a highly symmetric structure. Its assignment function $a(r_e)$ and characteristic angle $\alpha(r_e)$ take on forms where λ primarily influences the curvature and overall scaling, while the geometric pattern of lines in the Poincaré disk model (dictated by $\alpha(r_e)$) becomes canonical. This model provides a fundamental and analytically tractable example of an AEG space, serving as a valuable reference for more complex scenarios.